	CHAIN OF CUSTODY FORM (Original Stays With Samples)			US EPA Experimental Stream Facility 1003 US Highway 50, Milford, Ohio 45150 Phone # 513-965-4805 FAX # 513-965-4806					Originator:	
									Ship Date: 20180314	Contact:
									Destination: Cincinnati	Phone #:
									Recipient: Sam Smith	E-Mail:
									Phone #: 513-569-7681	
Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required	
20180312	16:20:00				1	DI	BTL BLK 1		TNH4, TRP, TNO2-3, TNO2, TN, TP	
20180312	16:23:00				1	DI	BTL BLK 2		"	
20180312	16:41:00				1	DI	Field BLK		"	
20180313	09:45:00	FL	42	0.1	1	SW	UKN		"	
20180313	10:20:00	FL	44	0.1	1	SW	UKN		"	
20180313	11:15:00	FL	43	0.1	1	SW	UKN		"	
20180313	11:50:00	FL	41	0.1	1	SW	UKN		"	
20180313	12:48:00	FL	4	0.1	1	SW	UKN		"	
20180313	13:04:00	FL	3	0.1	1	SW	UKN		"	
20180313	13:18:00	FL	14	0.1	1	SW	UKN		"	
20180313	13:42:00	FL	15	0.1	1	SW	UKN		"	
20180313	14:05:00	FL	7	0.1	1	SW	UKN		"	
20180313	14:35:00	FL	1	0.1	1	SW	UKN		"	
20180313	15:25:00	FL	5	0.1	1	SW	UKN		"	
20180313	15:38:00	FL	2	0.1	1	SW	UKN		"	
20180314	12:00:00	F			1	DI	HOUSE BLK		"	
20180312	16:20:00				1	DI	BTL BLK	2MHCl	TOC	
20180312	16:23:00				1	DI	BTL BLK	"	"	
20180312	16:41:00				1	DI	field BLK	"	"	
20180313	09:45:00	FL	42	0.1	1	SW	UKN	"	"	
20180313	10:20:00	FL	44	0.1	1	SW	UKN	"	"	
20180313	11:15:00	FL	43	0.1	1	SW	UKN	"	"	
20180313	11:50:00	FL	41	0.1	1	SW	UKN	"	"	
20180313	12:48:00	FL	4	0.1	1	SW	UKN	"	"	
20180313	13:04:00	FL	3	0.1	1	SW	UKN	"	"	
20180313	13:18:00	FL	14	0.1	1	SW	UKN	"	"	
20180313	13:42:00	FL	15	0.1	1	SW	UKN	"	"	
20180313	14:05:00	FL	7	0.1	1	SW	UKN	"	"	
20180313	14:35:00	FL	1	0.1	1	SW	UKN	"	"	
20180313	15:25:00	FL	5	0.1	1	SW	UKN	"	"	
	Relinquished By:		Date:	Time:	Received By:		Date:		Time:	

CHAIN OF CUSTODY FORM

(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:
Ship Date: 20180314 | Contact:
Destination: Cincinnati | Phone #:
Recipient: Sara Smith | E-Mail:
Phone #: 513-569-7681

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required
20180313	15:38:00	FL	2	01	1	SW	UKN	2N HCl	TOC
20180314	12:00:00				1	DI	HOUSE BLK	2N HCl	TOC

Relinquished By:	Date:	Time:	Received By:	Date:	Time:

CHAIN OF CUSTODY FORM

(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:
Ship Date: 20180319
Destination: Cincinnati
Recipient: Jim Smith
Phone #: 513-569-7681

Contact:
Phone #:
E-Mail:

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)
20180312	16:41:00				1	DI	FieldBlk	Chloro	372.44
20180313	09:45:00	FL	42	0.1	1	SW	UKN	"	123.92
20180313	10:20:00	FL	44	0.1	1	SW	UKN	"	204.26
20180313	11:15:00	FL	43	0.1	1	SW	UKN	"	171.29
20180313	11:50:00	FL	41	0.1	1	SW	UKN	"	173.28
20180313	12:48:00	FL	04	0.1	1	SW	UKN	"	154.55
20180313	13:04:00	FL	03	0.1	1	SW	UKN	"	211.35
20180313	13:18:00	FL	14	0.1	1	SW	UKN	"	216.46
20180313	13:42:00	FL	15	0.1	1	SW	UKN	"	224.53
20180313	14:05:00	FL	07	0.1	1	SW	UKN	"	258.12
20180313	14:35:00	FL	01	0.1	1	SW	UKN	"	267.17
20180313	15:25:00	FL	05	0.1	1	SW	UKN	"	322.48
20180313	15:38:00	FL	02	0.1	1	SW	UKN	"	263.99

Relinquished By:	Date:	Time:	Received By:

GAS TRAP DATA SHEET

Deployed 3/13/18 Retrieved 3/14/18

Lake:						
Site	**Lat**	**Long**	**Deployment time/date**	**Retrieval time/date**	**Volume recovered (mL)**	**Exetainer codes**
53	36.064185	78.744442	09:30	08:30	1	
42	36.065195	78.741936	~~09:43~~ 8:45	08:40	0	
50	36.068123	78.785996	10:06	08:52	0	
44	36.072775	78.774199	10:20	09:03	5	
47	36.068450	78.768252	10:30	09:36	2	
56	36.072118	78.766254	10:41	09:28	0	
52	36.070713	78.760339	10:50	09:18	0	
43	36.061535	78.773131	11:15	09:47	1	
54	36.057283	78.764926	11:25	10:00	0	
55	36.052962	78.761447	11:30	10:05	0	
41	36.055683	78.752171	11:50	10:14	0	
49	36.051040	78.756804	12:04	15:41	1	
57	36.048417	78.752456	12:15	10:26:00	0	
45	36.044087	78.752234	12:25	10:39:00	0	
10	36.039805	78.756057	12:31	10:47:00	0	
4	36.032188	78.761016	12:48	11:00:00	0	
8	36.029072	78.748451	12:57	11:10:00	0	
3	36.032455	78.736934	13:04	11:23:00	0	
14	36.029625	78.716916	13:18	11:45:00	1	
15	36.022647	78.714499	13:42	12:06:00	11	
11	36.019393	78.716959	13:48	12:15:00	9	
7	36.025578	78.668031	14:05	12:36	3	
1	36.020280	78.634384	14:35	12:50	0	
21	36.008872	78.688157	14:41	13:00	1	
13	36.033650	78.646322	15:00	13:15	0	
12	35.965655	78.633842	15:17	13:29	0	
5	35.970175	78.621171	15:25	13:42	0	
2	35.976232	78.588997	15:38	13:56	27	
16	35.956583	78.590546	15:55	14:15	0	
24	35.942223	78.582141	16:05	14:26	0	

DISSOLVED GAS AND AIR DATA SHEET

Lake:					Date:			
¥Site	Dissolved gas Exetainer codes		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes			BP (mm Hg)
42	42.DG.1	42.DG.2	115/25	4.4				
44	44.DG.1	44.DG.2		4.8				
43	43.DG.1	43.DG.2		4.3				
41	41.DG.1	41.DG.2		4.0	41.AA.1	41.AA.2	41.AA.3	41.AA.4
45	45.DG.1	45.DG.2		8.0				41.AA.5
04	04.DG.1	04.DG.2		9.0				
03	03.DG.1	03.DG.2		10.0				
14	14.DG.1	14.DG.2		7.0				
15	15.DG.1	15.DG.2		7.9				
07	07.DG.1	07.DG.2		7.8				
01	01.DG.1	01.DG.2		7.7				
05	05.DG.1	05.DG.2		7.9				
02	02.DG.1	02.DG.2		8.3	02.AA.1	02.AA.2	02.AA.3	02.AA.4

12:18:43 LGR Ambient

AA.5
14:06 LGR Ambient

¥Site: all dissolved gas samples collected at 0.1m depth

Boat motor should be off for 2 minutes before sample collection

Water Vol/Air Vol :: 115ml/25ml

Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES	
Lake:	FISH LAKE
Sample Date:	3/4/18
GGA file	SSA_2018-03-14_fo~1.txt * 10003.txt
gps file name	

DEPLOYMENT DATA 12:17:00 BADGER 13:16:30 LGR → clocks Windy

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	Water Temperature (C)	CH_4 and CO_2 profile checked
53	08:30:10	N	Y	50 45	6.8	Y
42	08:39:34	N	Y	50 ↑	6.4	Y
50	08:52:11	N	Y	50	6.7	Y
44	09:02:43	N	Y	50	6.8	Y
52	09:18:10	N	Y	50	7.6	Y
56	09:28:30	N	Y	50	7.8	Y
47	09:35:51	N	Y	50	7.4	Y
43	09:47:11	N	Y	50	7.5	Y
54	09:59:03	N	Y	50	7.3	Y
55	10:06:04	N	Y	50	7.3	Y
41	10:14:07	N	Y	50	6.9	Y
57	10:26:29	N	Y	50	7.1	Y
45	10:38:45	N	Y	50	8.8	Y
49	10:41:20	N	Y	45	9.0	Y
10	10:47:50	N	Y	50	9.1	Y
4	10:59:50	N	Y	50	8.9	Y
8	11:10:30	N	Y	50	9.5	Y
3	11:23:38	yes	Y	50	10.2	Y
3	11:26:47	N	Y	50	10.2	Y
14	11:45:47	N	Y	50	10.2	Y
15	12:06:44	N	Y	50	9.9	Y
11	12:14:56	N	Y	50	10.0	Y
7	12:36:30	N	Y	50	10.0	Y
1	12:50:47	N	Y	50	10.8	Y
21	12:59:30	N	Y	50	11.0	Y
13	13:13:35	N	Y	50	10.8	Y
12	13:28:42	N	Y	50 45	10.3	Y
5	13:42:40	N	Y	45	10.9	Y
2	13:56:17	N	Y	45	9.3	Y
16	14:14:39	N	Y	45	10.2	Y
24	14:26:10	N	Y	45	10.7	Y

SONDE DATA SHEET

Lake: Falls Sample Date: 3/12/18

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
53	0.8	0.1	5.82	11.76	93.8	0.253	8.75	12.0	26.7	
42	1.0	0.1	4.47	12.0	92.2	0.237	7.99	11.7	41.0	
50	1.0	0.1	5.75	12.12	96.1	0.130	7.86	9.0	33.6	
44	1.2	0.1	6.22	12.04	97.2	0.112	8.03	5.4	24.3	
47	2.0	0.1	6.48	11.85	96.2	0.163	8.00	8.8	22.5	
56	2.4	0.1	7.17	12.23	100.8	0.164	7.82	8.1	18.7	
52	1.2	0.1	6.17	11.54	92.8	0.151	8.01	11.8	17.0	
43	1.7	0.1	6.76	11.8	96.4	0.161	7.99	8.5	26.0	
54	1.8	0.1	6.32	11.87	96.3	0.156	8.0	12.1	25.0	
55	2.4	0.1	6.86	11.82	97.1	0.165	7.87	12.6	20.5	
41	0-8	0.1	6.82	12.14	99.6	0.163	7.84	11.8	24.7	
49	2.6	0.1	6.81	11.93	98.0	0.165	7.78	13.4	21.7	
57	2.1	0.1	7.83	11.80	100.0	0.169	7.90	[illegible]98	14.4	
45	2.3	0.1	8.36	11.40	96.9	0.172	8.01	8.0	10.8	
10	1.4	0.1	8.51	11.74	100.2	0.171	7.99	11.0	10.7	
4	0.9	0.1	7.95	12.05	101.7	0.160	7.90	8.8	21.4	
8	3.8	0.1	8.24	11.97	101.6	0.171	7.89	8.8	10.8	
3	1.6	0.1	8.79	11.62	100.2	0.164	7.90	12.5	8.3	
14	1.7	0.1	8.47	11.84	102.1	0.146	8.01	8.9	8.6	
15	4.2	0.1	8.47	12.00	102.1	0.157	7.81	8.5	8.3	
11	3.0	0.1	8.72	11.56	99.2	0.152	7.95	8.9	7.9	

* Copy of the Data on Drive

Site	Site Depth	Sample Depth	Temp	DO mg/L	DO %	Spec Cond	pH	chlor	Turb
7	~~8.1~~ 7.1	0.1	9.61	11.21	96.5	0.121	7.94	17.6	6.9
		1	9.64	10.93	96.0	0.121	7.93	24.5	6.9
		2	9.61	10.95	96.0	0.121	7.91	26.9	6.7
		3	9.60	10.89	95.5	0.121	7.90	26.3	6.6
		4	9.60	10.81	94.9	0.121	7.88	26.3	6.9
		5	9.58	10.77	94.4	0.121	7.87	25.5	6.9
		6	9.50	10.65	93.3	0.120	7.86	~~24~~.3 (24.3)	7.5
		7	9.45	10.53	92.0	0.120	7.85	24.7	11.3
		8	—						
1	3.5	0.1	10.32	11.05	98.7	0.105	7.96	15.4	5.5
21	3.0	0.1	10.15	11.47	102.8	~~0.121~~ 0.096	7.95	6.7	4.8
13	3.5	0.1	10.94	10.24	93.2	0.106	7.85	6.4	5.4
12	3.5	0.1	10.54	11.17	100.2	0.099	7.87	14.6	5.6
5	4.3	0.1	11.34	10.94	100.2	0.096	7.89	10.0	5.5
2	3.6	0.1	9.41	11.75	103.0	0.091	7.95	12.4	7.2
16	4.5	0.1	11.1	11.17	101.5	0.092	7.94	~~8.0~~ 7.0	7.1
24	2.5	0.1	12.37	10.88	102.6	0.089	7.87	6.2	7.1
7 - revisit		0.1						12.0	
7		8.0						25.0	

→ Cleaned Sensors to confirm Chlor Fs

Date 03/14/18 Pre-deployment

Cal cylinder serial # SSA_2018-3-14-f0000.txt CC498857

Dilution system serial # EPA # A20808

LGR file

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.0180	0.17	06:17:10	06:20:00
9	60.0	2.0	20.2	1.8180	18.60	06:20:00	06:22:40
8.5	60.0	2.1	21.4	1.9218	19.70	06:22:40	06:24:54
8	60.0	2.2	22.7	2.0370	21.02	06:24:54	06:27:30
7.5	60.0	2.4	24.2	2.170	22.55	06:27:30	06:28:50
7	60.0	2.6	25.9	2.3230	23.97	06:28:50	06:30:16
6.5	60.0	2.7	27.9	2.5030	25.99	06:30:16	06:31:23
6	60.0	3.0	30.2	2.7110	28.32	06:31:23	06:32:43
5.5	60.0	3.2	32.9	2.958	30.80	06:33:26	06:34:40
5	60.0	3.6	36.1	3.250	34.00	06:34:40	06:35:50
		Room	Air	2.0000	445.5	06:10:00	06:17:10

Signature _____

Date 3/16/18 Post-deployment

Cal cylinder serial # CC4 98857

Dilution system serial # AZ0808

LGR file

959-2018-03-16_f0000.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.012	-0.01	13:23:50	13:25:50
9	60.0	2.0	20.2	1.810	18.34	13:20:45	13:22:30
8.5	60.0	2.1	21.4				
8	60.0	2.2	22.7	2.03	20.85	13:22:30	
7.5	60.0	2.4	24.2				
7	60.0	2.6	25.9	2.320	23.80	13:18:44	13:20:45
6.5	60.0	2.7	27.9	~~2.32~~	~~23.8~~ JN		
6	60.0	3.0	30.2				
5.5	60.0	3.2	32.9	2.950	30.65	13:17:00	13:18:44
5	60.0	3.6	36.1	3.250	33.70	13:14:50	13:17:00

Signature _____

Date 3/23/18
Cal cylinder serial # CC163816
Dilution system serial # EPA # A208 08
LGR file 23a-2018-03-23_f0000.txt
Sample MFC gas correction factor 1.43

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.014	0.00	14:50:00	14:53:15
7	70.0	28.3	283.2	25.50	261.6	14:53:15	14:58:00
6.5	70.0	30.5	304.7	27.60	284.4	14:58:00	~~14:~~ 15:00:45
6	70.0	33.0	329.8	29.90	309.0	15:00:45	15:03:11
5.5	70.0	35.9	359.4	32.55	338.2	~~15:07:22~~ ~~15:03:11~~	15:09:45
5	70.0	39.5	394.9	35.67	372.0	15:03:11	15:07:22
4.5	70.0	43.8	438.1	39.42	413.8	15:09:45	15:12:14
4	70.0	49.2	491.9	44.02	467.0	15:12:14	15:14:45
3.5	70.0	56.1	560.8	49.71	531.3	15:14:44	15:17:00
3	70.0	65.2	652.1	57.45	618.5	15:17:00	15:19:17
	Room	Air	Prior	to	Calibration		

Signature _____

ATTACHMENT A
YSI 6600 SONDE CALIBRATION LOG SHEET

SONDE ID: STD 16 **DATE:** 3/23/18 **TECHNICIAN:** JTW

DATE / TIME OK? YES / NO

BATTERY VOLTS: 12.41 (Change if < 10.0 V) **CHANGED BATTERIES?** YES / (NO)

TURBIDITY WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

CHLOROPHYLL WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

DISSOLVED OXYGEN: "DO Sat %" **BAROMETRIC PRESSURE:** 754.6 mm Hg

SENSOR OUTPUT TEST: PASS / FAIL (For Rapid-Pulse Probes, change membrane if failed)

Record first 10 readings from Discrete Sampling:

_____ _____ _____ _____ _____

_____ _____ _____ _____ _____

DO MEMBRANE CHANGED? YES / NO **PROBE RE-SURFACED?** YES / NO **DATE/TIME:** _____

[~97] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 99.8
DO CHARGE [50 ± 25]: _____ **DO GAIN** [1.0 ± 0.35]: _____

DEPTH: "Pressure" [~~m~~ ft] **PRESSURE OFFSET** [VENTED = 0 ± 6 / NON-VENTED = -14.7 ± 6]: _____
[0] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** -0.365

CONDUCTIVITY: "SpCond" [mS/cm] **CONDUCTIVITY CELL CONSTANT** [5.0 ± 0.45]: _____

[10] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 1.399 22 °C

TURBIDITY: "Turb+NTU" [NTU]

[0] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 0.2

[~~1000~~ 124] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 121

CHLOROPHYLL: "Chlor ug/L" [ug/L]

[0] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** -0.1

~~[113]~~ **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 11.8 5.5 site 26 site 43

---

pH: "pH"

[7] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 6.9 **mV** [0 ± 40]: _____

[10] **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 9.9 **mV** [-177 ± 40]: _____

ORP: "ORP mV" **ORP mV OFFSET [0 ± 100]:** _____

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